

3D PRINTER FILAMENT EXTRUDER MACHINE



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FINAL YEAR PROJECT REPORT (SESSION 2020-2024)



**DEPARTMENT OF MECHATRONICS & BIOMEDICAL
ENGINEERING**

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Abstract

In recent times, 3D printing has become the go to method of prototyping among engineers but this method of prototyping has added to the persistent problem of plastic waste. This plastic waste from 3D printed prototyping can be negated by recycling wasted 3d printing filament and making it a new. The goal of the project is to provide a better alternative to 3D printing companies than to import expensive 3D printing filament and waste residual filament left behind by printing. This 3D Printer Filament Extruder Mechanism would provide a low cost and recyclable method to make 3d printing more lucrative and inexpensive. This Project will describe the design process, including the selection of materials, the choice of components, and the assembly of the extruder. This project will demonstrate the feasibility and effectiveness of producing 3D printer filament using a custom-built extruder. The extruder has the potential to significantly reduce the cost of 3D printing by producing high-quality filament at a lower cost than commercially available filaments.

Nomenclature

Symbol	Abbreviation
ABS	Acrylonitrile Butadiene Styrene
°C	Degree Centigrade/Celsius
DLP	Digital Light Processing
FDM	Fused Deposition Modeling
FYP	Final Year Project
PLA	Polylactic Acid
SLA	Stereolithography
USA	United States of America

Table of Contents

1	Introduction	1
1.1	Background and Motivation	2
1.1.1	Sustainable Development Goals	2
1.2	Problem Statement.....	2
1.3	Project Objective.....	2
1.4	Scope.....	2
1.5	Flowchart	3
1.6	Block Diagram.....	3
1.7	Cost Analysis	4
1.8	Timeline of the Project.....	5
2	Literature Review	9
3	Modeling And Simulation	12
3.1	Mathematical Modeling:	12
3.3.1	@Solidworks Simulation Of the Base Of Mechanism:	20
4	Hardware and Modular Working.....	24
4.1	Hardware Descriptions:	24
5	Discussion and Result.....	30
5.1	Discussion:	30
5.1.1	Temperature Control Of Filament:	30
6	Conclusion and Future Recommendation.....	33
6.1	Conclusion:	33
6.2	Future Recommendation:	33
	Reference	
	Appendixes	

List of Figures

Figure 1 Filabot Desktop Filament Extrusion Solution.....	9
Figure 2 Wellzoom desktop filament extruder.....	9
Figure 3 Flowchart for Filament Extruder Mechanism.....	11
Figure 4 Mechanism of 3D printer Extruder.....	18
Figure 5 Barrel, screw and heat band.....	21
Figure 6 Result of the Mathematically Calculation.....	24
Figure 7 Simulink of DC Gear Motor.....	26
Figure 8 Graph show the angular acceleration and velocity.....	27
Figure 9 Graph show the speed before and after gear ratio.....	27
Figure 10 Graph show the Speed of the DC gear motor.....	28
Figure 11 Base of mechanism on Solidworks.....	29
Figure 12 Maximum Force applied (167N)	30
Figure 13 Result at required force (45N)	30
Figure 14 Direction of Force.....	31
Figure 16 Coupling of the motor.....	33
Figure 17 Thermocouple will attached the extruder.....	34
Figure 18 Spooling System.....	34
Figure 19 DC gear motor.....	35
Figure 20 Screw.....	35
Figure 21 Barrel heater band.....	36
Figure 22 Temperature Controller.....	36
Figure 23 Thermocouple.....	37
Figure 24 Inline Filament Diameter Sensor.....	37

List of Table

Table 1 Table of Cost	3
Table 2 Table of Project Division.....	6
Table 3 Temperature Control of filament.....	22

1 Introduction

Due to its simplicity in creating complicated designs, 3D printing has gained popularity as a manufacturing tool. For small firms and hobbyists in particular, the high cost of 3D printing filament can be a substantial entrance hurdle. Using a specially designed extruder to create filament is one way to solve this issue. To create filament for 3D printing, a filament extruder melts and extrudes plastic granules or pellets. This document shows the design and analysis of a filament extruder for 3D printers that was created to lower the price of filament for 3D printing.

A startup called Filabot focuses on making desktop filament extruders for 3D printing the filabot as shown in Figure 1. With the use of their goods, consumers may produce their own 3D printing filament from basic components like plastic pellets, offering greater customisation and cost savings. Based in Vermont, USA, Filabot was established in 2012. The Filabot Original, Filabot Wee, and Filabot EX2 are just a few of the filament extruder models that they sell. They also sell extras like spoolers and pellet grinders.



Figure 1 Filabot Desktop Filament Extrusion Solution

Wellzoom as mentioned below in Figure 2 is a Chinese brand that specializes in filament extruder development and manufacturing. They also produce a desktop solution.

1.1 Background and Motivation

A project to build a 3D printer filament extruder attempts to build a device that can make plastic pellets or recycled plastics into unique 3D printing filaments. This project was started with the intention of giving 3D printing fans, who frequently rely on pricey pre-made filaments, an economical and environmentally friendly answer. You may customize a filament's attributes, including color, tensile strength, and flexibility, as well as save waste by recycling plastics, by making your own filament. The chemistry, physics, and extrusion processes can also be studied using this project as a teaching aid.

1.1.1 Sustainable Development Goals

The Sustainable Development Goals as Organized by United Nations Department of Economic and Social Affairs That will be kept in mind while devising Our Objective for The Project are:

- **Goal#09:** Industry, Innovation and Infrastructure
- **Goal#12:** Responsible Consumption and Production
- **Goal#13:** Climate Action

1.2 Problem Statement

3D printer filaments are all imported to Pakistan and cost very much and the extruders that are available are very expensive as well. Goal is to design and manufacture a Low Cost with high quality output filament extruder.

1.3 Project Objective

To Design and Develop a Low-Cost 3D Printer Filament Extruder Mechanism

1.4 Scope

For this project, the scope is limited to just manufacturing filament with required diameter keeping suitable tolerance and have kept the whole system desktop friendly and compact.

1.5 Flowchart

Explaining the procedure of Filament Extruder Mechanism by flowchart as shown in Figure 3

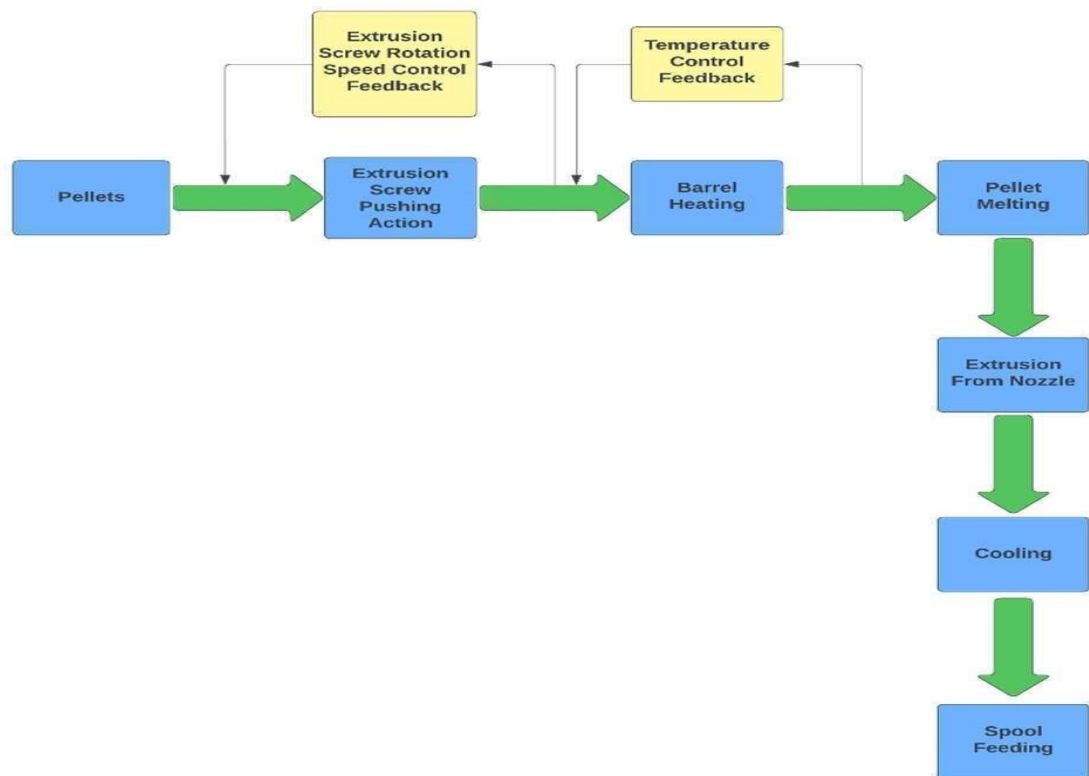


Figure 3 Flowchart for Filament Extruder Mechanism

1.6 Block Diagram

Draw an Extruder Electronic Control block diagram which shows the visual representation of a system that uses a simple, labeled block that represents the multiple items and concept as given below in Figure 4.

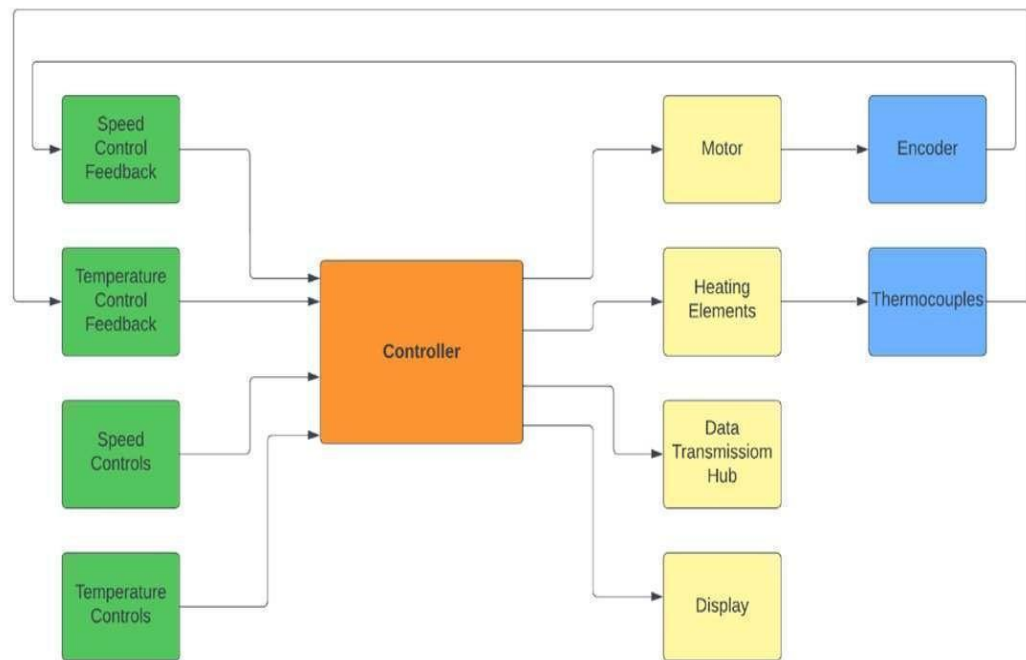


Figure 4 Initial Block Diagram for Extruder Electronic Control

1.7 Cost Analysis

In the below Table 1 ,Shows the prices of the components which are used in this project. This is the preliminary cost analysis of the project .

Table 1 Table of cost

Components	Prices
Screw	Self-Manufacturing: 15000-20000 Online Sourcing: 40000-55000
Barrel	10000
Electrical sensors	6000
Cooling fans	1000
Flow control sensor	500
Heating element	10000
Controller board	Arduino Mega
Power supply	TBD
Hopper	TBD
Nozzle	TBD
Frame	8000-14000
Filament spool holder	2500-3000

1.8 Timeline of the Project

Timeline as projected below in Figure 5, Figure 6 and Figure 7 explained the work to be completed in 3 phases.

FYP#01 (2023) Timeline

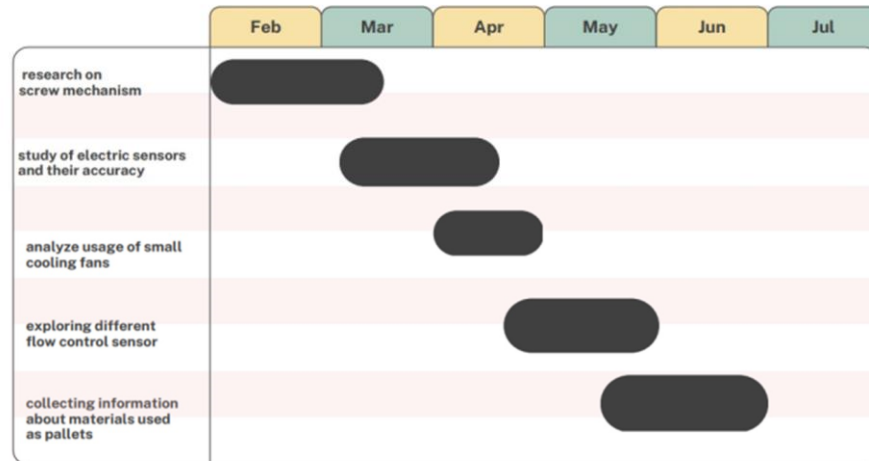


Figure 2 FYP1 Timeline

FYP#02 (2023-2024) Timeline

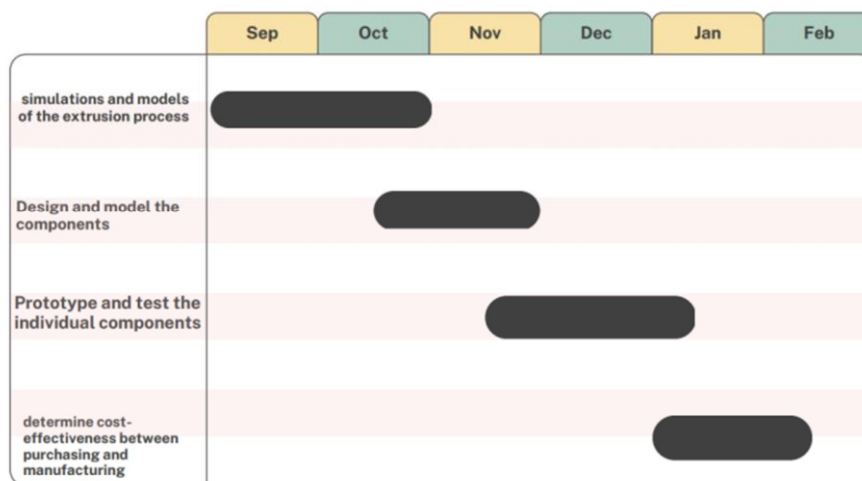


Figure 3 FYP2 Timeline

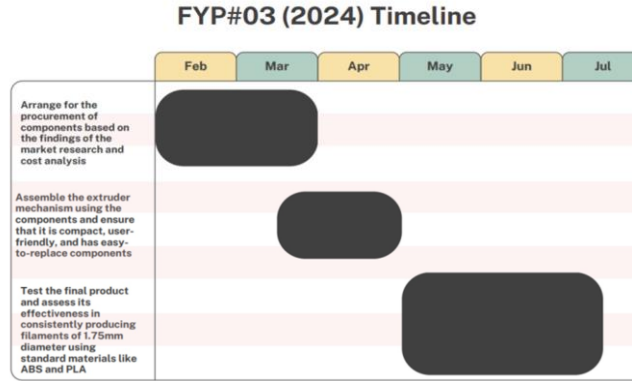


Figure 4 FYP 3 Timeline

1.9 Work Division

Table 2 shows the work distribution for final year project among group members.

Table 2 Project Division

Work Division	Subayyal Sajjad	Muhammad Umer	Fatima Nazir
Design selection	✓	✓	✓
Calculations			✓
Mathematical Modeling	✓	✓	✓
CAD Modelling	✓		
Software/ Coding		✓	
Components Selection		✓	✓
Fabrication	✓	✓	
Final Testing	✓	✓	✓
Report Writing	✓		✓

1.10 Organization of Report:

Chapter 1 Introduction: In This Chapter we go over the basics of This project. We discuss its Academic background, Related hardware implementations and Relevant research.

Chapter 2 Literature review: In this chapter we go over the literature review for the research of project. We discuss in the Methodology and we also discuss the Tentative Timelines and Cost Analysis for the Project in this section.

Chapter 3 Modeling and Simulation: In this chapter we go over the basic of the project is mathematical modeling and simulation. We doing the mathematically and simulation of DC gear motor and extruder. We also doing the simulation of base of the project on Solidworks.

Chapter 4 Hardware and Modular Working: In this chapter we go over the basic of this project. We discuss the hardware components which are used in the project.

Chapter 5 Result and Discussion:

In this chapter we go over the basic of the project results, outcome and discussion about the project.

Chapter 6 Conclusion:

In this chapter we conclude the project and discuss the improvement in the project in future.

2 Literature Review

Literature review is the heart and soul of any engineering project. Knowledge of pre-existent research on our topic of interest leads to a better understanding of the current state of an engineering problem and helps us in understanding the next step for a project.

Study carried out by Mona Et al. published in 2019, designed the components for the extruder e.g. Auger (Screw), Barrel, Hopper, Extrusion Nozzle etc. They also emphasized the importance of material selection of extrusion nozzle that it should be able to withstand temperatures of 150-200°C. They were vague on heating element selection and Motor type selection but provided the starting base for the Extrusion screw design [1].

This project by Vaibhav Et al. specifically incorporates for existing design of making plastics filament extruder for 3D printing machine. Plastics filament extruder produces plastic filament of different types of diameters by using the dies. In the 3D printer, extruder is responsible for pushing the correct amount of filament to the hot end where it is then melted and extruded.

In Their Project For the making of plastic filament extruder used the pellet and granules in the form of input materials pellets. These materials have a melting point below 200 degrees and hence it can be controlled by using temperature control. They used the ABS pellets for the plastic filament extruder. During plastic filament extrusion, their main focus was on creating diameter and melting point of filament from ABS pellets. Theoretical analysis was also conducted by them to estimate the required parameters diameter of screw, length of screw width of screw, volume of hopper, weight & band heater, availability of components and their costs. Then these theoretical results were compared with the experimental results available; thus, validating the results. For the making of filament extruder that can be used by small scale manufacturing units [2].

Aminur 'Ilman Et Al, in their paper argued that 3D printing is a growing technology that has found applications in various industries, including aerospace and medical device manufacturing. However, failed prints can generate waste,

which has led to a growing interest in using recycled filament to reduce waste and make 3D printing more sustainable. Cost-effectiveness is achieved through the simplicity of processes used, such as FDM, SLA, and DLP. Filament types include PLA and ABS. The Filastruder extruder machine can recycle 3D printing waste into useful filament by melting pellets in the extruder barrel, forcing them through a die, and cooling them onto a spool. Tensile strength comparisons have shown that the original ABS material outperforms recycled ABS material in different mechanical properties [3].

Nassar Et Al, In Their Study emphasized about the FDM which is fused Deposition modeling which is used in a 3D printing process that uses a continuous filament of a plastic material show in the figure [2]. Actually the working of filament in FDM is fed from a large spool through a moving heated printer extruder head, and finished on the growing work [4].



Figure 4 Mechanism of 3D printer Extruder

By rely Et Al, studied the emissions caused by Filament extruders and their effects on the surrounding. They also measured the amount of emissions and recorded them in the form of Figures and charts.[5].

The 3D printer filament extruder machine based on Fuzzy Logic Controller by Pribadi Et Al. which was powered by solar cells is able to recycle plastic waste into high-quality products. It uses AI to maintain temperature and speed control of the DC motor and is indeed an environment friendly product [6].

Hachimi Et Al studied Thermoplastic polymers that can be formed by injection molding, extrusion, or 3D printing and have a wide range of industrial applications. Making filaments for 3D printing out of scrap plastic, pellets, and granules is one way to reduce the price and waste of these resources [7]

The literature analysis concludes by highlighting the diverse uses of 3D printing filament extrusion and highlighting a move towards sustainability, material innovation, experimental validation, technological integration, and industrial applications.

3 Modeling And Simulation

In chapter 3 The modeling and simulating of 3D printer filament extruder mechanisms, Understanding the intricate procedures involved with this cutting-edge technology is important. Using different software for modeling and simulation methods, in this way we know how these mechanisms work and perform. This helps them make 3D printing more able to work or operate quickly and introduce new methods.

Basically, this is a 3d printer filament extruder model in which the focus is on the mathematical modeling of screw, barrel and extruder.

3.1 Mathematical Modeling:

The mathematical modeling DC gear motor , extruder, barrel and nozzle. In the process of extrusion use the material of PLA and ABS. Making molten plastic mass out of solid plastic pellets is the main job of the plastic extrusion machine. This mass may then be used to make products of any size or shape. Among the many product categories in which this equipment finds use are pipes, different types of extrusions, bottles (blow molding), and injection moldings.

The barrel modeling includes the representation of the filament contact and barrel surface, the barrel's size and design, the flow of material and its melting behavior inside the barrel, the temperature distribution along the barrel, and the barrel's design. Perhaps part of the modeling process involves analyzing the shear and pressure forces acting on the filament as it goes through the barrel.

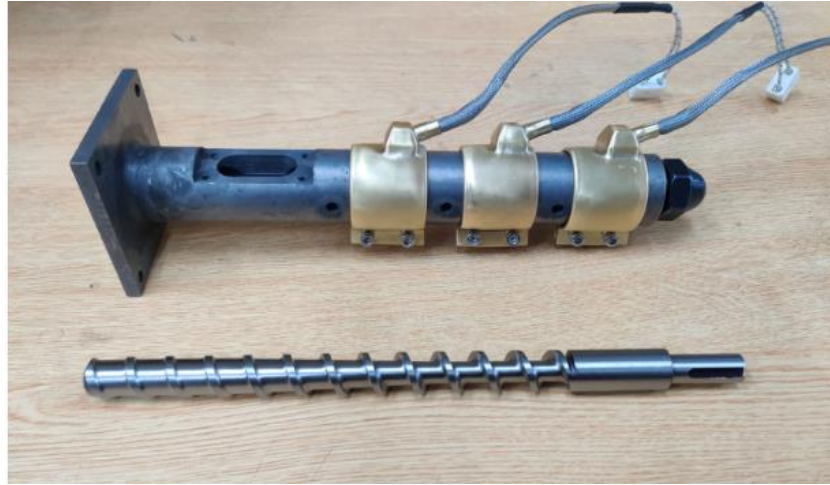


Figure 5: Barrel, screw and heat band

Mathematical modeling of the nozzle, barrel, and extruder [8] on MATLAB consists of parameters which are used in the mechanism of the 3D printer filament extruder mechanism. Now show the mathematical modeling on the MATLAB.

3.2 Mathematical Calculation

The chapter entails the determination and calculations of the various components of the machine adopted in this research. The single screw extruder machine was designed to have a capacity of extruding 1200cm^3 of polymers.

3.2.1 Feed Zone Calculation:

The geometry of the feed zone of the screw is given by the following data:

Barrel Diameter $D_b = 20\text{mm}$

Screw flight width $w_{\text{flight}} = 3\text{mm}$

Channel width $w_c = w_e = 13\text{mm}$

Depth of the feed zone $H_{fz} = 6\text{mm}$

Screw speed $N_2 = 60\text{rpm}$

Conveying efficiency $\eta_f = 0.35$

Bulk density of the polymers $= 1200\text{ kg/m}^3$

Helix angle $\varphi = ?$

We using the formula:

$$\tan\varphi = \frac{\text{pitch}}{\pi D_s} \quad (1)$$

Screw diameter = 20mm

$$\tan\varphi = \frac{16}{\pi \times 20\text{mm}}$$

$$\varphi = 14.2866^\circ \quad (2)$$

3.2.2 Solid Conveying Rate In The Feed Zone:

The solid conveying rate in the feed zone of the extruder can be calculated as:

$$G = 60 \times p_o \times N_2 \times n_f \times \pi^2 \times H_{fs} \times D_b (D_b - H_{fs}) \left(\frac{w_e}{w_e + w_{flt}} \right) \times \sin\varphi \cos\varphi$$

Putting the value then we get it

$$G = 6.495 \times 10^{-5}. \quad (3)$$

3.2.3 Analysis the Flow in the Extruder:

1. Drag Flow Q_d

We find the drag flow by using the formula of drag flow. The pressure distribution of the flow in the extruder is the total output obtained from the drag flow, back pressure flow and leakage. Assuming that there is no leakage.

$$Q_d = \frac{1}{2} \pi^2 \times N_2 \times \pi^2 \times H_{cd} \times D_s^2 \times \sin\varphi \cos\varphi \quad (4)$$

$$Q_d = 0.0016994 \text{ m}^3 / \text{s}.$$

$$Q_{max} = Q_d = 0.0016994 \text{ m}^3 / \text{s}. \quad (5)$$

2. Pressure Flow

When there is no pressure build up at the end of the extruder, any flow is due to drag and maximum flow rate Q_{max} can be obtained. The equation then can be reduced to only the drag term as follows

$$Q_p = \frac{\pi \times D_s \times H_{cd}^3 \times \sin\varphi}{12^n} \times \frac{P_a}{L_{esl}} \quad (6)$$

The L_{esl} is the screw length = 226.5 mm

For finding the flow rate of pressure flow we find the P_a and η . First we find the η by using the formula for the η .

Flow Viscosity $\mu = 200 \text{ Pa.s}$

Consistency Index **m = 10779.53 Pa.s**

Power Law Index **n= 0.45**

Bulk Density **$\rho = 1200 \text{ Kg/m}^3$**

The power law is usually represented as $\eta = m\dot{\gamma}^{n-1}$, where m is sometimes replaced by 'k' or other letter. The consistency index is said to include the temperature dependence of the viscosity whilst the power law index represents the shear thinning behavior of the polymer melt. "The limits of the law are zero (0) and infinity".

Now for the η we using the formula is:

$$\eta = m\dot{\gamma}^{n-1} \quad (7)$$

Now as we know that the $\dot{\gamma}$ is defined as :

$$\dot{\gamma} = \frac{6Q}{H_{FLT}^2 w_{FLT}} \quad (8)$$

Now putting value in the equation (6)

$$\dot{\gamma} = 9266.6 \text{ s}^{-1}. \quad (9)$$

Now putting the value of (7) in the equation (5)

$$\eta = m\dot{\gamma}^{n-1} \quad (10)$$

$$\eta = 19.7853 \text{ Pa.s.} \quad (11)$$

Now we find the P_a we using the formula which is given as below:

$$P_a = \frac{6\pi \times L_{esl} \times \eta \times N_z^n \times D_2}{H_{cd}^2 \times \tan\phi}$$

$$P_a = 11624.1915 \text{ Pa.}$$

Now putting the value in the equation

$$Q_p = \frac{\pi \times D_s \times H_{cd}^3 \times \sin\phi}{12 \times \eta} \times \frac{P_a}{L_{esl}} \quad (12)$$

Then the pressure flow rate is:

$$Q_p = 7.24 \times 10^{-4} \text{ m}^3/\text{s}.$$

3.3 Simulation Of DC Gear Motor:

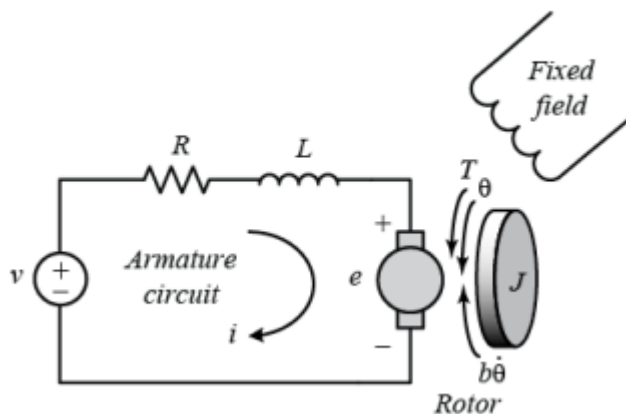
Using a parameter, the model of a DC gear motor display its speed, angular acceleration, angular velocity, and gear ratio. determined the value of the

parameter employed in the theoretical DC motor calculation. Next, use the value of the parameter that is needed to calculate the DC gear motor to display the DC gear motor Simulink. High torque is needed to extrude plastic along the barrel and Filament extrusion machine into the dye hole. The motor must rotate slowly and steadily during the procedure. Reasonably sized servo motors and stepper motors are unable to provide adequate torque to maintain speed. Conversely, a desktop-sized geared DC motor that satisfies all of the above requirements is available. Industrial DC motors that are the proper size and prepared for application.



Figure 6 Result of the Mathematically Calculation

Mathematical Calculation:



The torque generated by a DC motor is proportional to the armature current and the strength of the magnetic field. t the motor torque is proportional to only the

armature current (I) by a constant factor motor torque constant, K_t as shown in the equation below.

$$\mathbf{T} = \mathbf{K}_t \cdot \mathbf{I} \quad (1)$$

The back emf, e , is proportional to the angular velocity of the shaft by a constant factor back-emf constant, K_e .

$$\mathbf{e} = \mathbf{K}_e \dot{\theta} \quad (2)$$

The motor torque and back emf constants are equal, that is,

$$\mathbf{K}_t = \mathbf{K}_e \quad (3)$$

Therefore, K is used to represent both the motor torque constant and the back emf constant.

Now apply the Kirchhoff's voltage law from given figure.

$$\mathbf{J}\ddot{\theta} + \mathbf{b}\dot{\theta} = \mathbf{K}\mathbf{i} \quad (4)$$

$$\mathbf{L}\frac{\partial i}{\partial t} + \mathbf{R}\mathbf{i} = \mathbf{V} - \mathbf{K}\dot{\theta} \quad (5)$$

Applying the Laplace transform, the above modeling equations can be expressed in terms of the Laplace variable s .

$$\mathbf{s}(\mathbf{J}\mathbf{s} + \mathbf{b})\theta(\mathbf{s}) = \mathbf{K}\mathbf{I}(\mathbf{s}) \quad (6)$$

$$(\mathbf{L}\mathbf{s} + \mathbf{R})\mathbf{I}(\mathbf{s}) = \mathbf{V}(\mathbf{s}) - \mathbf{s}\theta(\mathbf{s}) \quad (7)$$

By removing $\mathbf{I}(\mathbf{s})$ from the two a for mentioned equations, we obtain the following open-loop transfer function, where the rotational speed is regarded as the output and

The voltage of the armature is regarded as the input.

$$\mathbf{T}(\mathbf{s}) = \frac{\mathbf{w}(\mathbf{s})}{\mathbf{V}(\mathbf{s})} = \frac{\mathbf{K}}{(\mathbf{J}\mathbf{s} + \mathbf{b})(\mathbf{L}\mathbf{s} + \mathbf{R}) + \mathbf{K}^2} \frac{\mathbf{rad/sec}}{\mathbf{V}} \quad (8)$$

Now the transfer function of the gear motor. Frist we find the gear ratio is G is:

$$\mathbf{G} = \frac{\text{Output gear}}{\text{input gear}} \quad (9)$$

$$\mathbf{G} = \frac{48}{13}$$

G is gear ratio which has no unit.

$$T(s) = \frac{\frac{K}{G}}{LS*J + (LS*b + R*Js) + (R*b + K^2)} \frac{rad/sec}{V} \quad (10)$$

By concerning the datasheet of the motor the simulink model is designed and simulated

Simulation:

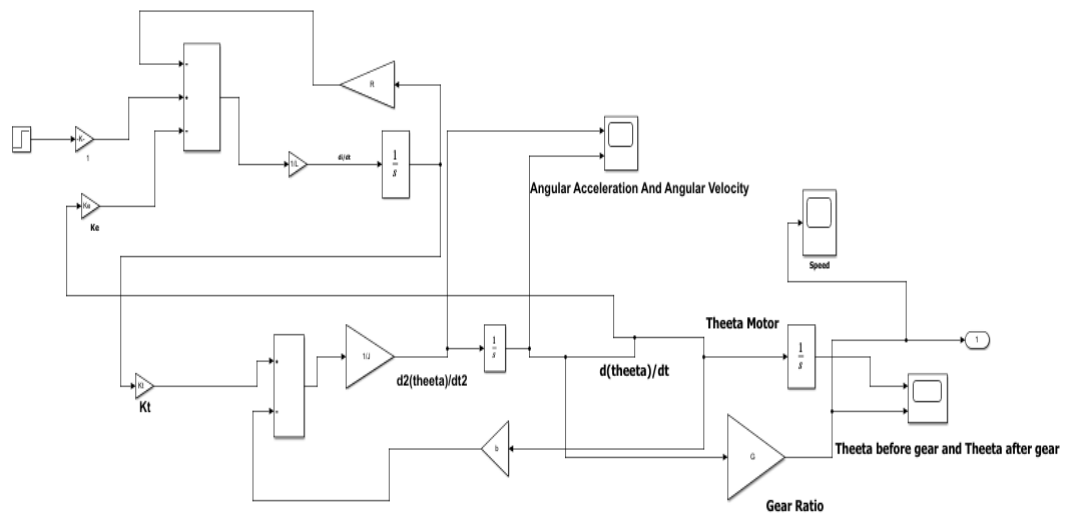


Figure 7 Simulink of DC Gear Motor

Result:

In the result the graph shoe the relationship of the angularat acceleration and angular velocity To graph the relationship between angular acceleration and angular velocity, first determine if the system's dynamics are linear or non-linear and identify any forces such as constant torque or damping.

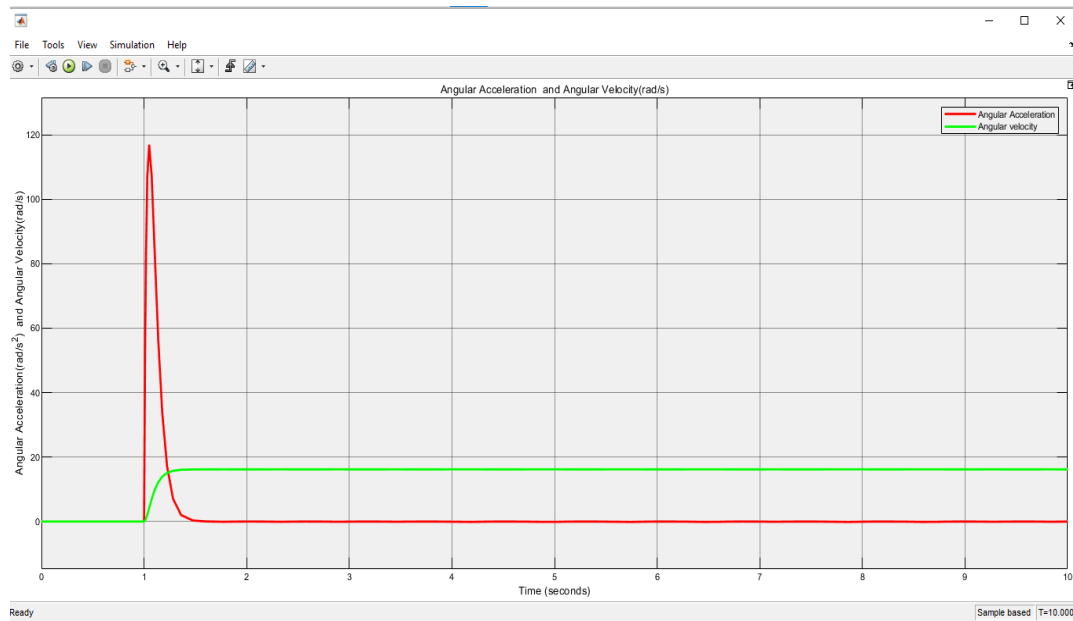


Figure 8 Graph show the angular acceleration and velocity

this graph show the relationship of gear ratio. A gear ratio affects the relationship between input and output speed in a gear system. Before applying a gear ratio, the speed is higher but the torque is lower. After applying a gear ratio, the output speed decreases while the torque increases proportionally.

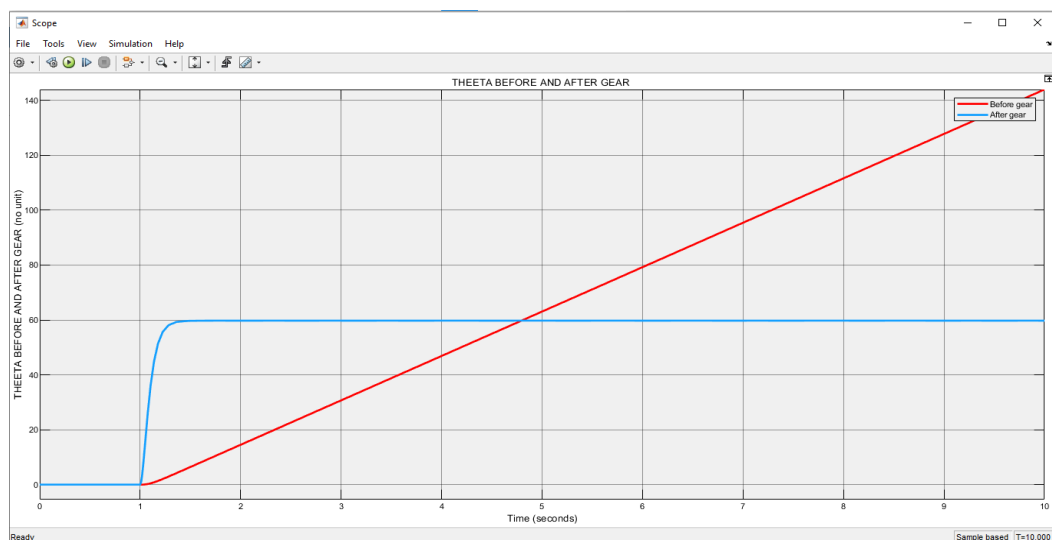


Figure 9 : Graph show the speed before and after gear ratio

In this graph show the relationship of speed and time and determine the no-load speed and the stall torque of the motor. Measure the motor speed at different applied voltages or loads.

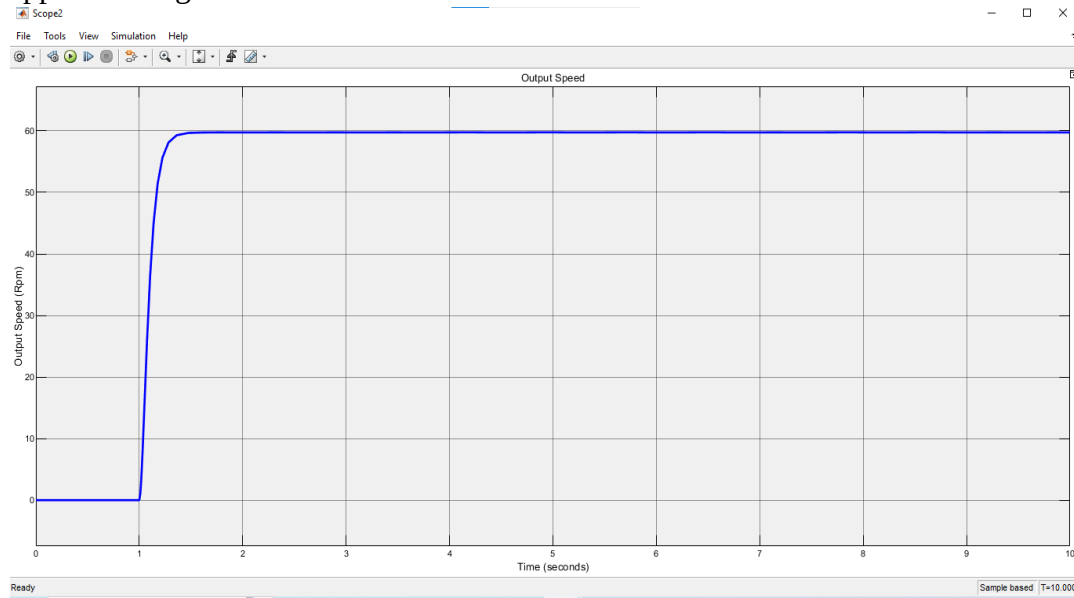


Figure 10 Graph show the Speed of the DC gear motor

3.3.1 @Solidworks Simulation Of the Base Of Mechanism:

This simulation the failure of the base of extruder against the force applied by the motor. The motor using is a DC geared motor and provides a maximum torque of about a 30Nm. Then rotate the mechanism of the extruder by using a Chain + Sprocket at a distance of 0.18m. So, done the simulation by putting the force applied by the motor in the two possible directions towards the shaft it is going to move.

Calculation of Force:

Required force as the required torque is 8Nm at max so at a displacement of 0.18m force will be...

$$F = T/r$$

$$F = 8/0.18$$

$$F = 44.44.N$$

So, in the two directions we applied 23N, 23N forces and found out there will be no failure and there will be no permanent deformation as well.

Max Force:

The max torque the motor can provide is about 30Nm, So at a displacement of 0.18m force will be

$$F=T/r$$

$$F=30/0.18$$

$$F=166.66N$$

So, in the two directions applied 83N,83N forces and found out there will be no failure and there will be no permanent deformation as well.

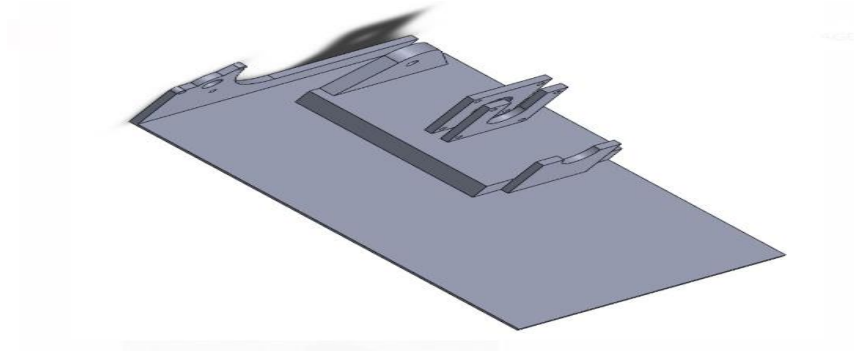


Figure 11 Base of mechanism on Solidworks

Simulation:

According to von Mises stress theory, material yields when a critical distortion value is reached. Our simulation on SOLIDWORKS also support the stability of the structure and claims that there will be no failure as per the von mises failure criteria

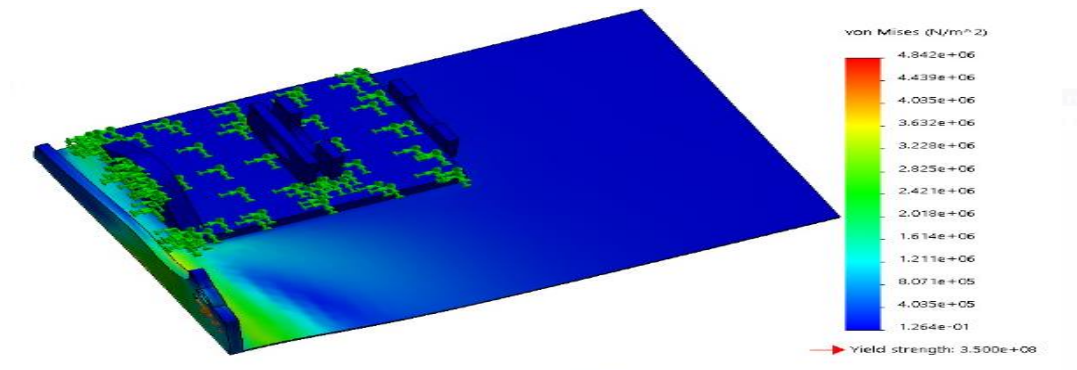


Figure 12 Maximum Force applied (167N)

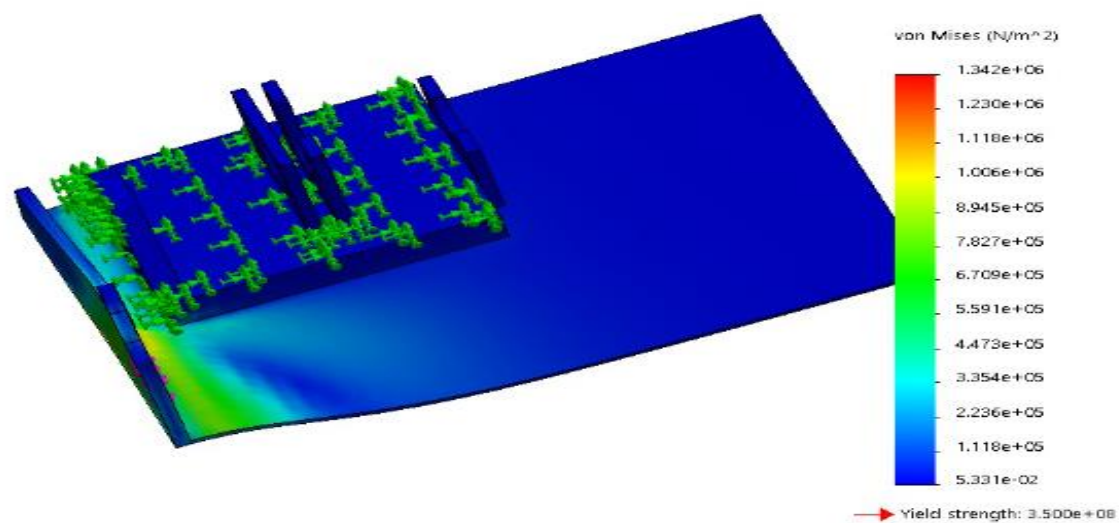


Figure 13 Result at required force (45N)

In this simulation the failure of the base of our extruder against the force applied by the motor the motor we are using is a DC geared motor and provides a maximum torque of about a 30Nm.

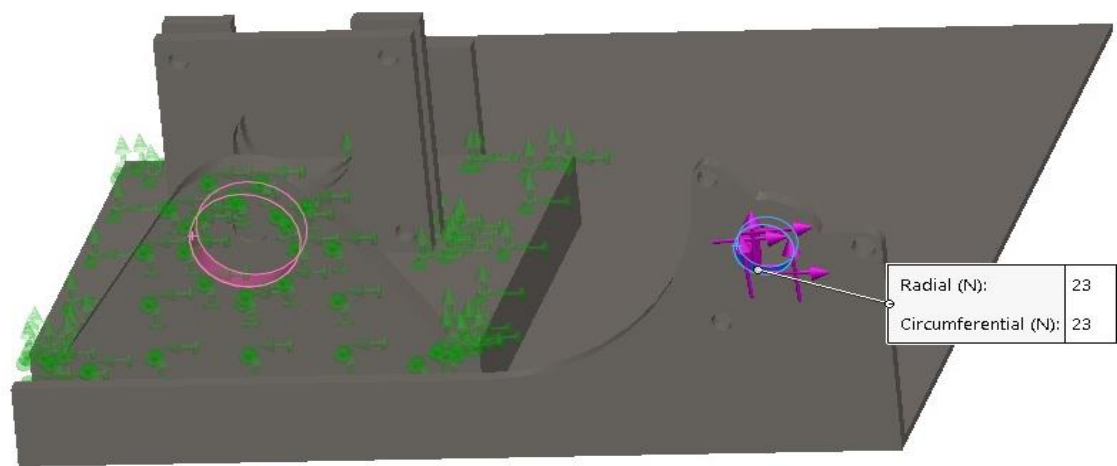


Figure 14 Direction of Force

The 3D printer filament extrusion mechanisms were thoroughly examined in this chapter using MATLAB and SolidWorks for mathematical modeling and simulations. To ensure effective operation and correct functioning, comprehensive assessments included barrel dynamics, nozzle parameters, and DC gear motor .

simulations. The structural integrity and suitability of the base for the extrusion process were confirmed by means of simulations conducted on SolidWorks, which verified its stability against applied forces.

4 Hardware and Modular Working

4.1 Hardware Descriptions:

Hardware components are fully attached to the extruder. Brief description of hardware assembly is given below. One of the most important parts of a 3D printer is the filament extruder mechanism, which melts and extrudes thermoplastic material to produce the filament needed for printing. The following is a brief overview of its hardware components:

- **Extruder Nozzle:**

This is where the molten filament is deposited onto the print bed. It typically has a small diameter hole through which the melted filament is extruded.

- **Heating Element:**

Usually a high-temperature resistor or heater cartridge, this component heats up to melt the solid thermoplastic material. It's often located close to the filament feed mechanism.

- **Filament Feed Mechanism:**

This includes a gear or wheel that grips the filament and feeds it into the extruder. It may also have a motor to drive the filament forward.

- **Barrel:**

This is the part where the filament is melted. It consists of a metal tube (barrel) with the heating element and nozzle attached. The barrel heats up to melt the filament, and the nozzle directs the molten plastic onto the print bed.

- **Cooling System:**

Some extruders incorporate a cooling fan or other cooling mechanism to prevent heat from traveling too far up the filament, which could cause unwanted melting and deformation in the upper part of the extruder.

- **Drive System:**

The filament is forced into the extruder by this device. The arrangement, in which the motor is placed far away and drives the filament through a tube to the hoted, or a direct-drive system, in which the motor is positioned next to the extruder, might be used.

- **Mounting Hardware:**

Various nuts, bolts, brackets, and other hardware are used to mount the extruder securely to the 3D printer frame.



Figure 16 Coupling of the motor



Figure 17 Thermocouple will attached the extruder

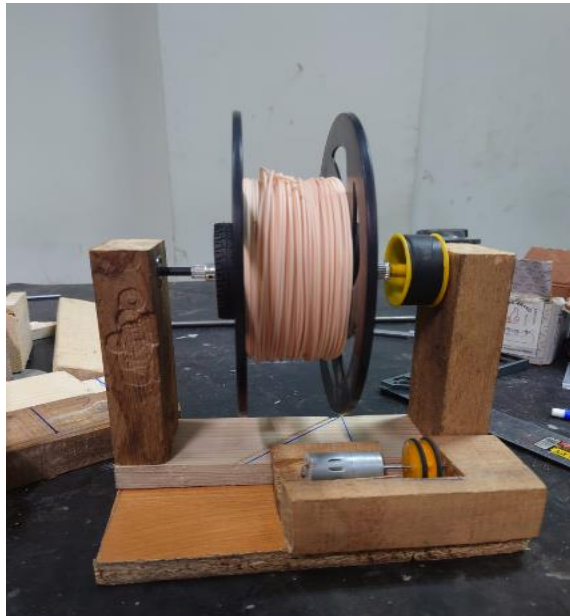


Figure 18 Spooling System

4.2 Components:

The filament extrusion mechanism of a 3D printer typically divided into several components which is explain in the detailed form:

DC Gear Motor:

A motor needs to move slowly and steadily with a strong twisting force in order to push plastic through a hole with a machine. Conventional motors, such as servomotors or stepper motors, aren't fast enough or strong enough for this. However, a smaller geared DC motor—such as the ones in the Figure 13 automobile windshield—works well since it provides sufficient force and a range of speed options. Large industrial DC motors that meet the requirements are costly and require specialized machinery to operate.



Figure 19 : DC gear motor

The Barrel-Auger-Die Assembly

Screw:

The barrel at the center of the plastic extrusion system directs molten plastic toward a die, which aids in creating plastic filament with the necessary dimensions. To force the plastic pellets forward and develop pressure toward the die, a propulsion system must be mounted inside the barrel. Melted plastic is forced out of the die's hole once the necessary pressure is produced.



Figure 20 Screw

Barrel with the Heat Band:

The use of heat bands wrapped around the barrel is crucial for effectively melting plastic for extrusion. These rings precisely raise the barrel's temperature so that the plastic softens without decline. Accurate shaping and pushing of the plastic through the die is ensured by this regulated heat. The plastic can attain the proper viscosity for extrusion because the heat bands keep the barrel's length at a constant

temperature. In the absence of this controlled heat, the plastic may not melt evenly, which could cause problems throughout the extrusion process.

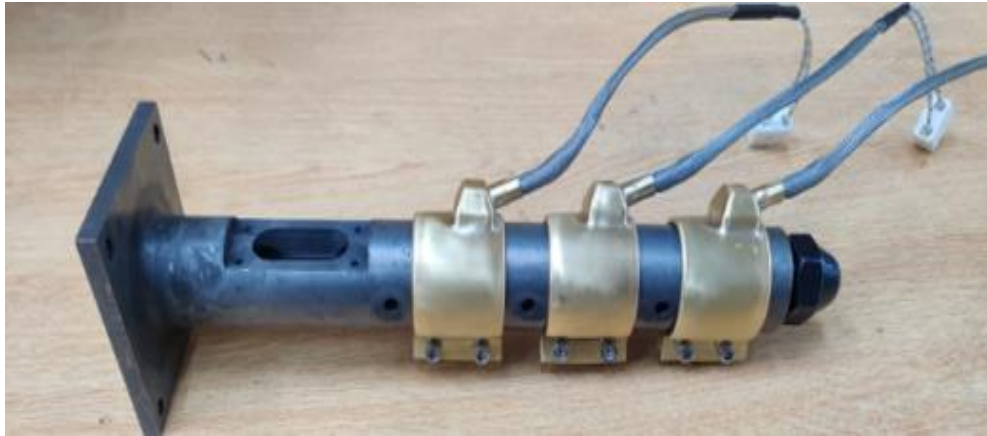


Figure 21 Barrel heater band

Temperature Controller:

A temperature controller is a piece of equipment used to regulate and uphold a particular system temperature. In order to maintain the atmosphere at the intended setpoint, it continuously checks the temperature and modifies the heating or cooling components.



Figure 22 Temperature Controller

Thermocouple:

A thermocouple is used as a temperature sensor in filament extrusion to gauge the temperature within the barrel or extrusion chamber where the plastic is melted. To

ensure accurate control of the heat bands or heating elements, this sensor measures temperature variations and transmits the information to a temperature controller.

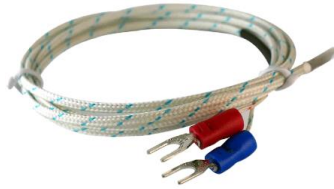


Figure 23 Thermocouple

Inline Filament Diameter Sensor:

The sensor measures the width of 3D printing filament. It sends real-time data to the printer's control system. The system adjusts printing settings based on this data. This ensures consistent extrusion during printing. Ultimately, it helps create accurate and reliable 3D prints.



Figure 24 Inline Filament Diameter Sensor

This chapter focuses on crucial hardware elements for efficient 3D printer filament extrusion. It highlights the significance of a geared DC motor, the barrel-screw assembly, and heat bands for precise plastic melting and shaping. Additionally, it emphasizes temperature control tools like controllers and thermocouples, essential for maintaining optimal extrusion conditions. This chapter offers key insights into hardware essentials, ensuring a controlled and effective filament extrusion process in 3D printing.

5 Discussion and Result

5.1 Discussion:

In this section discuss the presents experimental result on the properties of composite filament for 3D printer. The chapter includes a through analysis and discussion of the results obtained, including the important data, Interpretation of the results, and comparison to other research studies. This chapter is organized into several key sections, which is visual observation, the diameter of filament and effect of changing extrusion speed on filament quality. The results of the FYP will depend on various factors, including the specific goals that is set, the equipment and materials that is use, and how we conduct the experiments and analysis. Here are some results that might expect from a project.

Result:

5.1.1 *Temperature Control Of Filament:*

Different filament materials have different options of extrusion temperatures. The material ABS which is used for the filament extrusion. ABS material required the temperatures closer to 220°C -240°C. At the 200°C temperature, the ABS filament doesn't melt enough, making prints weak with bad layer sticking and gaps between layers. The prints look rough and not nice. At 230°C temperature is perfect for ABS filament. It melts smoothly and flows evenly, resulting in stronger prints with layers that stick together well.

Table 3 Temperature Control of Filament

Controller	Temperature	P	I	D
C1	180°	50	40	60
C2	240°	20	240	60

The surface of the prints looks nice and smooth, and there's less chance of them getting warped. At 240° improves filament flow and reduces nozzle blockages, but excessive heat can cause stringing and surface flaws, necessitating careful temperature monitoring.

5.1.2 *Effect of Changing Extrusion Speed on Filament Quality:*

When we adjust how fast the material comes out in a 3D printer filament-making project, it affects how good the filament turns out. Going faster makes more filament, but it might not be as consistent or strong. Going slower makes the filament more even and strong, but it takes longer to make. So, finding the right speed is important to make good filament without wasting time.



Figure 25 Extrusion of filament

5.1.3 *Initial Filament Extrusion Test: Observing Brittle Results:*

Start assembling the 3D printer extrusion setup, leaving out the fan cooling system and spool. Utilized the important components such as the motor, thermocouple, relay, and extruder screw. Start the extrusion process, the extrusion process, melting the ABS polymer filament material without cooling, the filament quickly cools down post-extrusion. Insufficient cooling prevents the material from properly hardening, making it fragile.

Observing that the filament is to break or change shape, which means it's not good for printing. Emphasize the significance of integrating proper cooling systems to produce strong and practical 3D printing filaments.



Figure 26 Final filament Extrusion

6 Conclusion and Future Recommendation

6.1 Conclusion:

In the conclusion, the project involved design and construction of a working prototype with the constraints being that it had to be portable and easy to operate. In order to achieve this goal by creating a cost-effective 3D printer filament extruder mechanism. The resulting filament, measuring 1.75 millimeters in diameter, in a line with the standard size used in 3D printing applications. This success makes sure that the filament works with many different 3D printers out there. By achieving these objectives, the project contributes to making 3D printing more accessible and affordable for makers and small businesses. Additionally, the successful development of the extruder. The mechanism shows creativity and engineering skill in overcoming obstacles related to making filament. Overall, the project's results have an opportunity to improve 3D printing technology's safety and availability. To design and manufacture a Low Cost with high quality output filament extruder.

6.2 Future Recommendation:

In the current system, the user must ensure that sufficient quantity of pellets is present in the hopper. A monitoring system can be implemented to monitor and inform the user about the quantity of pellets present in the hopper. The current design for the prototype has the capability to produce plastic filaments with 1.75mm diameter. To change this diameter, the end cap with the extrusion hole has to be changed out with another end cap with the required hole size. It is suggested that further extrusion process optimization be taken seriously to improve uniformity and cost. Also, studying possible improvements or changes to the extrusion mechanism may result in increased reliability and efficiency.

Conclusion

In the current system the weight of machine is heavy by using the material is Steel and aluminum. To change the material of machine is carbon fiber and fiberglass which the strength and the durability is less than the steel and aluminum.

Now in the future adjust how fast and hot it works to use less energy but make more filament. Use heaters and coolers to save energy while still making filament fast.

Try lighter materials for the machine parts to use less energy and make the machine lighter. Use the different heaters according to the material of filament temperature range .

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Appendices

Mathematically modeling of Motor on Matlab:

Clc

clear all

J = 0.00052 %Moment of inertia of the rotor (kg.m²)

b = 0.000115% Viscous friction coefficient (N.m.s)

Gi=13% Input gear

Go=48 %Output gear

R = 8 %Armature resistance (ohms)

L = 0.2 %Armature inductance (henries)

Mathematically Calculation of Flow Rate:

% Given parameters

Db = 0.02 % Barrel Diameter (m)

wf = 0.003 % Screw flight width (m)

wc = 0.013 % Channel width (m)

we = 0.013 % Channel width (m)

Hfz = 0.006 % Depth of the feed zone (m)

Hcd = 0.06

N2 = 60 % Screw speed (rpm)

nf = 0.35 % Conveying efficiency

Pitch = 0.016 % Pitch (m)

Sd = 0.02 % Screw diameter (m)

L_esl = 0.2265 % The L_esl is the screw length (m)

mu = 200 % Flow Viscosity (Pa.s)

m = 10779.53 % Consistency Index (Pa.s)

n = 0.45 % Power Law Index

rho = 1200 % Bulk Density (Kg/m³)

Ds = 0.02 %screw diameter

phi = atan2d(Pitch, pi * Db);

% Displaying the value of ?

```

disp(['? : ', num2str(phi)]);
%Solid Conveying Rate In The Feed Zone
% Calculating the value of sin(?) and cos(?)
sin_phi = sind(phi)
cos_phi = cosd(phi)
% Calculating G using the provided formula
G = 60 * Pitch * N2 * nf * pi^2 * Hfz * Db * (Db - Hfz) * (we / (we + wf)) *
sin_phi * cos_phi;
% Displaying the value of G
disp(['Value of G: ', num2str(G)]);
%Analysis the Flow in the Extruder
%Drag Flow Q_d
% Drag Flow
Q_d = 0.5 * pi^2 * N2 * Hcd * Ds^2 * sin_phi * cos_phi;
% Displaying the value of Q_d
disp(['Value of Q_d: ', num2str(Q_d)]);
% Calculating ?
gamma = 6 * Q_d / (Hfz^2 * wf);
% Displaying the value of ?
disp(['Value of ?: ', num2str(gamma)]);
% Calculating ?
eta = m * gamma^(n - 1);
% Displaying the value of ?
disp(['Value of ?: ', num2str(eta)]);
% Calculating Pa
L_esl
eta
N2
n
Ds
Hcd

```

```

phi
tan(phi)
Pa = ((6*pi)*L_esl*eta*(N2^n)*Ds)/((Hcd^2)*tand(phi))
% Displaying the value of Pa
disp(['Value of Pa: ', num2str(Pa)]);
% Calculating Q_p
Q_p = (pi * Ds * Hcd^3 * sind (phi)) / (12 * eta) * Pa / L_esl;
% Displaying the value of Q_p
disp(['Value of Q_p: ', num2str(Q_p)]);

```

OUTPUT:

```
>> CALCULATION OF EXTRUTION
```

```
Db =
```

```
0.0200
```

```
wf =
```

```
0.0030
```

```
wc =
```

```
0.0130
```

```
we =
```

```
0.0130
```

```
Hfz =
```

```
0.0060
```

```
Hcd =
```

```
0.0600
```

N2 =

60

nf =

0.3500

Pitch =

0.0160

Sd =

0.0200

L_esl =

0.2265

mu =

200

m =

1.0780e+04

n =

0.4500

rho =

1200

Ds =

0.0200

φ : 14.2936

sin_phi =

0.2469

cos_phi =

0.9690

Value of G: 6.4913e-05

Value of Q_d: 0.0016984

Value of γ : 94355.2714

Value of η : 19.7915

L_esl =

0.2265

eta =

19.7915

N2 =

60

n =

0.4500

Ds =

0.0200

Hcd =

0.0600

phi =

14.2936

ans =

-6.3420

Pa =

1.1624e+04

Value of Pa: 11624.1915

Value of Q_p: 0.00072369

Features

Miniature construction
Low-Noise Output
4.5 V to 6 V Operation
Magnetically Optimized Package
Linear output for circuit design flexibility
Temperature range of -40 °C to 85 °C



3 pin SIP (suffix UA)

Description

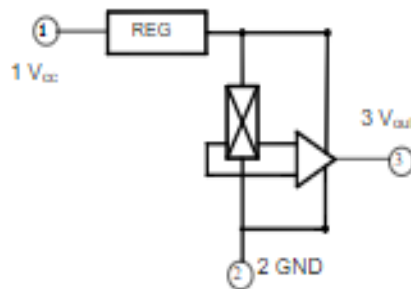
SS49E Linear Hall-effect sensor is small, versatile linear Hall-effect device that is operated by the magnetic field from a permanent magnet or an electromagnet. The linear sourcing output voltage is set by the supply voltage and varies in proportion to the strength of the magnetic field. The integrated circuitry features low

noise output, which makes it unnecessary to use external filtering. It also includes thin film resistors to provide increased temperature stability and accuracy. The linear Hall sensor has an operating temperature range of -40 °C to 85 °C appropriate for commercial, consumer and industrial environments.

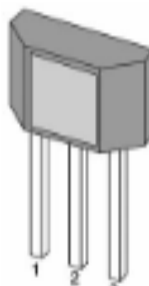
Typical Applications

Motor control
Magnetic code reading
Ferrous metal detector
Current sensing
Position sensing

Functional Block Diagram



Pin Definitions and Descriptions



Name	No	Status	Description
Vdd	1	P	Power Supply
Gnd	2	P	IC Ground
Output	3	O	Output

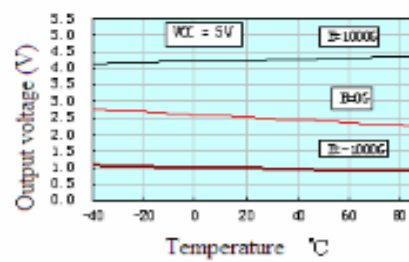
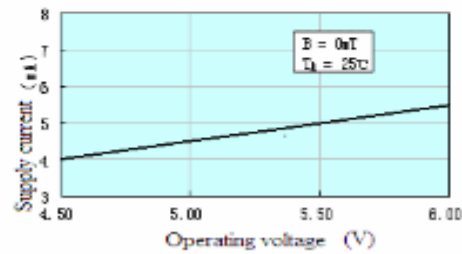
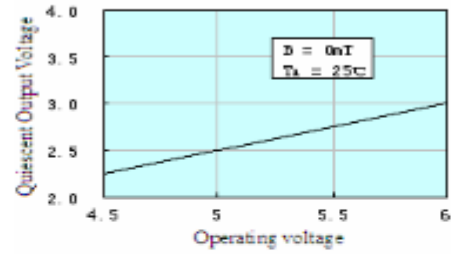
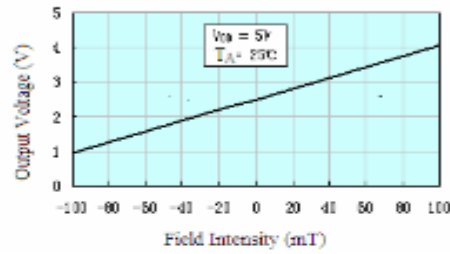
Absolute Maximum Ratings

Parameter	Symbol	Value	Units
Supply Voltage (operating)	V_{CC}	8.0	V
Output Current	I_{OUT}	20	mA
Operating Temperature Range	T_A	-40~85	°C
Storage Temperature Rang	T_S	-65~150	°C

Electrical Characteristics ($T_A=25^{\circ}\text{C}$, $V_{CC}=5.0\text{V}$)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Units
Operating voltage	V_{CC}	Operating	3.0		6.5	V
Supply current	I_{CC}	Average		4.2	8.0	mA
Output Current	I_{OUT}		1.0	1.5		mA
Response Time	T_{ack}			3		μs
Quiescent Output Voltage	V_o	$B=0\text{G}$	2.25	2.5	2.75	V
Sensitivity	ΔV_{out}	$T_A=25^{\circ}\text{C}$	1.6	1.8	2.0	mV/G
Min Output Voltage		$B=-1500\text{G}$		0.86		V
Max Output Voltage		$B=1500\text{G}$		4.21		V

Performance Characteristics



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